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it for the first time, the display will show 'New PWD:' message and wait for the six-digit password to be entered. The password (except 000000, which is invalid) will be saved to the EEPROM and displayed on the LCD. Any password from 000001 to 999999 may be used.

Now, the display will show 'Open PWD:' message and wait for the six-digit password. If the password matches with the saved password, the relay will energise for ten seconds and 'Lock Opened' message will appear on LCD1.

After ten seconds, relay RL1 will de-energise and the display will wait for the password again.

In case, the password does not match with the saved password, the display will show 'Invalid' and wait for the correct password.

A 5V DC power supply (300mA or above) is required to operate the circuit.

The firmware code (PasswordLock.c) for ATtiny2313 MCU is written in C language. Hex code can be generated using any AVR C compiler such as Programmers Notepad (WinAVR).

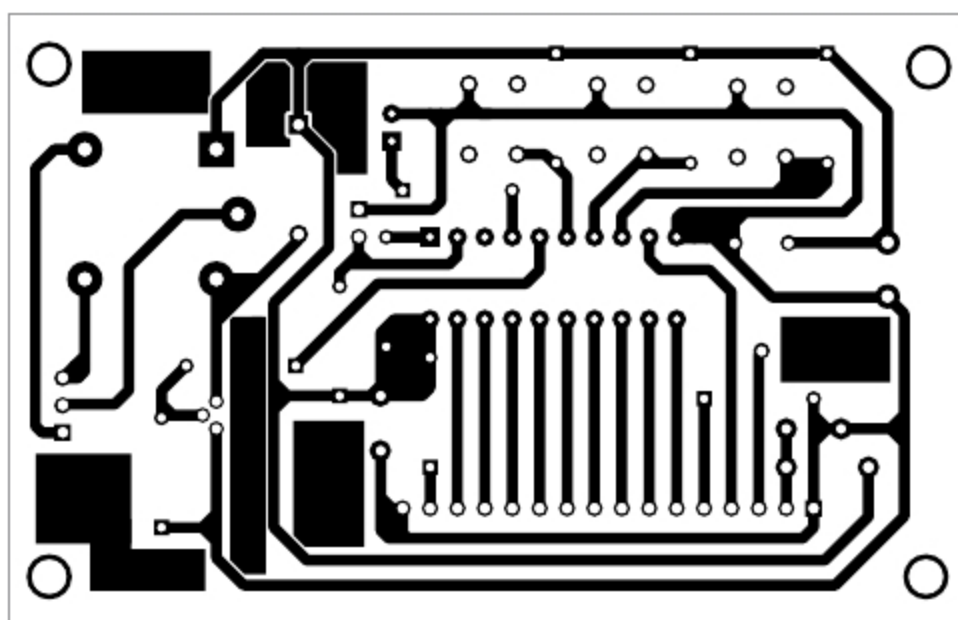


Fig. 2: Actual-size PCB layout of the password locker

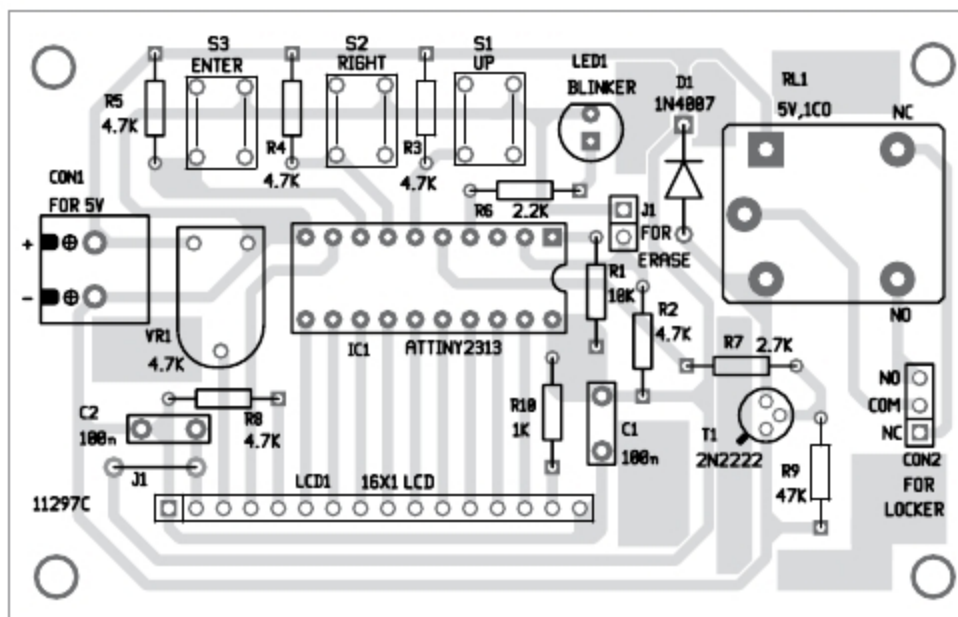


Fig. 3: Components layout for the PCB

Construction and testing

An actual-size PCB layout of the password locker is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB or veroboard, program/burn ATtiny2313 with passwordlock.hex file using AVR programmer.

Connect 5V DC power supply across connector CON1. For the first time, you will have to set a password as explained above. The blinker LED1 blinks while waiting for password input. If the password is correct, RL1 will energise and the lock connected across the relay contacts will open. The LED1 will now glow continuously, indicating the circuit is ready to use. **EFY**



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EFY Note
The source code of this project is available for free download at source.efymag.com